

Question Paper

Exam Date & Time: 29-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

DIGITAL SYSTEM DESIGN [DS_2123]

Marks: 50

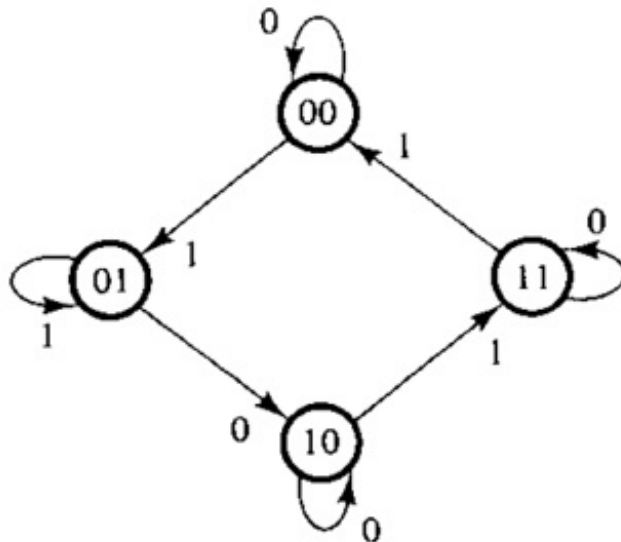
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the complete synchronous sequential circuit design procedure to realize the functionality described by a given state diagram. The final circuit must be implemented using JK Flip-Flops. (5)

A)



Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Design a 2-input OR gate using static CMOS logic. Illustrate the primary sources of power dissipation in the resulting CMOS circuit. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Analyze the given Boolean function
 $f(x_1, x_2, x_3, x_4) = x_1'x_3'x_4' + x_3x_4 + x_1'x_2'x_4 + x_1x_2x_3'x_4$ with the don't-care set $D = \Sigma(9, 12, 14)$ to find the minimum-cost SOP and POS expressions for the function. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) Design the least cost implementation with 2:1 multiplexer and other necessary gates for the following function using Shannon's expansion: (5)

A) $F(a, b, c, d) = abc + a'cd' + bcd$

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Implement a three-input XOR gate using 4-to-1 multiplexer with a neat diagram. (3)

- C) The circuit that has four input signals, x_1 , x_2 , x_3 , and x_4 , and three output signals, f , g , and h . It implements the logic functions (2)
- $$g = x_1x_3 + x_2x_4$$
- $$h = (x_1 + x_3')(x_2' + x_4)$$
- $$f = g + h$$
- Implement this design in Verilog HDL.

- 3) Design and illustrate 4-bit shift register with the parallel load that operates according to the following function table: (5)

- A) **Shift Load Register Operation**

0	0	Shift left
0	1	Load parallel data
1	0	Complement
1	1	No change

- B) Determine the phenomenon of "race-around" that can occur in a simple edge-triggered JK Flip-Flop. Explain how the architecture of a Master/Slave Flip-Flop prevents this problem by effectively isolating the input from the output during the critical clock transition. (3)

- C) Examine the JK flip-flop with a logic diagram, truth table, a characteristic equation and an excitation table. (2)

- 4) Since a computer has many registers, paths must be provided to transfer information from one register to another. An efficient scheme for transferring the information between registers in a multiple-register configuration is a common bus system. Design a 4-bit common bus system for 4-registers of 4-bits each using multiplexers. (5)
- A)

- B) Determine the key structural difference between a Synchronous Counter and an Asynchronous Counter. Justify why the synchronous approach, where all flip-flops are clocked simultaneously, is essential for high-speed operation in digital systems. (3)

- C) Distinguish between blocking and non-blocking assignments in Verilog with an example. (2)

- 5) Analyse the use of Multiplexers to implement a bus. Describe the primary components of this bus structure and apply this structure to explain how data selection is performed (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Design a 3-bit Johnson counter by implementing the following steps. (3)

- a) Draw the state-transition diagram.
b) Construct a state table using T flip-flops.
c) Design the Counter.

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Determine the number of unique states generated by a k=5 bit Johnson Counter. Write the general formula relating the number of states to the number of flip-flops k. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_2123:DIGITAL SYSTEM DESIGN

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	10
3	3	CO 3	5
4	4	CO 4	20
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_2123:DIGITAL SYSTEM DESIGN

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	13	43(86%)
4	Analysis	2	7(14%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)